SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: **Database Management Systems**
COURSE CODE: **CSA0501**

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity.* (BL2, BL3)

Activity Tool : Experiments (High)
Assessment Tool Weightage: 40%

QUESTIONS Total Marks: 50

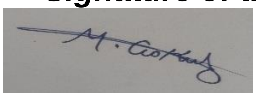
Q.No	Question	Bloom's Level	Marks
1	For the relation STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade), identify all functional dependencies and determine the candidate keys.	BL2 – Understand	**5**
2	Demonstrate the process of converting an unnormalized relation into 1NF with step-by-step justification and example.	BL3 – Apply	**5**
3	Given relation R(A,B,C,D,E) with FDs: A→BC, BC→D, D→E. Analyze and decompose it to 2NF eliminating partial dependencies.	BL3 – Apply	**5**
4	Apply 3NF decomposition on the relation EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID) with transitive dependencies.	BL3 – Apply	**5**
5	Verify whether R(A,B,C,D) with FDs: AB→C, C→D, D→A is in BCNF. If not, decompose it into BCNF.	BL3 – Apply	**5**
6	Experimentally verify lossless-join decomposition for a given relation using the tabular (Chase) algorithm.	BL3 – Apply	**5**
7	Identify the multi-valued dependencies in TEACHER(TID, Subject, Hobby) and decompose it into 4NF.	BL3 – Apply	**5**

8	Demonstrate how join dependencies lead to redundancy. Decompose a given relation to 5NF (Project-Join Normal Form).	BL3 – Apply	**5**
9	Given a poorly designed database schema for a college, identify all anomalies (insertion, deletion, update) and normalize it to 3NF.	BL2 – Understand	**5**
10	Compute the closure of attribute sets for the given FDs and determine all candidate keys for the relation.	BL3 – Apply	**5**

Code of Conduct:

I M . G O K U L   1 9 2 5 6 5 0 6 7 _ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results

Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Q1. STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade) — FDs and candidate key

FDs:

- SID → SName
- CID → CName, Instructor
- SID, CID → Grade

Closure of {SID, CID}: adds SName, CName, Instructor, Grade → all attributes. No proper subset works (SID alone can't give CName/Instructor/Grade; CID alone can't give SName/Grade).

Candidate Key = {SID, CID}

Q2. UNF → 1NF

A relation is in 1NF when every attribute holds a single atomic value (no repeating groups).

Steps:

1. Identify the relation with repeating/multi-valued attributes.
2. Expand the repeating group into separate rows.
3. Duplicate the atomic attributes for each new row.
4. Redefine the key so each row is unique.

Example — UNF:

SID	SName	Courses
S1	Arun	DBMS, OS, CN
S2	Divya	DBMS, Java

Converted to 1NF:

SID	SName	Course
S1	Arun	DBMS
S1	Arun	OS
S1	Arun	CN

SID	SName	Course
S2	Divya	DBMS
S2	Divya	Java

Key becomes {SID, Course}.

Q3. R(A,B,C,D,E), FDs: A→BC, BC→D, D→E — decompose to 2NF

Closure of {A}: A→BC gives B,C; BC→D gives D; D→E gives E → $\{A\}^+ = \{A,B,C,D,E\}$ = all attributes.

Candidate key = {A} (single attribute).

2NF is violated only by *partial* dependency on part of a *composite* key. Since the key is a single attribute, no partial dependency is possible.

Conclusion: R is already in 2NF.

(Note: it still has a transitive dependency D→E, which would need removing for 3NF: R1(A,B,C,D) and R2(D,E).)

Q4. EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID) — 3NF

FDs:

- EmpID → EmpName, DeptID, ManagerID
- DeptID → DeptName

Transitive dependency: EmpID → DeptID → DeptName (DeptID is non-prime) — violates 3NF.

3NF Decomposition:

- EMPLOYEE(EmpID, EmpName, DeptID, ManagerID)
- DEPARTMENT(DeptID, DeptName)

Q5. R(A,B,C,D), FDs: AB→C, C→D, D→A — verify/decompose to BCNF

Candidate keys (via closures): $\{AB\}^+=\{A,B,C,D\}$, $\{BC\}^+=\{A,B,C,D\}$, $\{BD\}^+=\{A,B,C,D\}$ → **AB, BC, BD** are all candidate keys. Every attribute is prime, so R is in 3NF.

BCNF check:

FD	LHS closure	Superkey?
AB→C	ABCD	Yes
C→D	ACD	No (missing B)
D→A	AD	No (missing B,C)

Not in BCNF. Decompose:

1. Using C→D: R1(A,C,D), R2(B,C)
2. R1 still violates BCNF (D→A, D not superkey) → split into R1a(A,D), R1b(C,D)

Final BCNF relations: R1a(A,D) key D; R1b(C,D) key C; R2(B,C) — all trivially in BCNF, lossless join.

Q6. Verify lossless-join with the Chase algorithm

Example: R(A,B,C,D), FD C→D, decomposed into R1(A,B,C) and R2(C,D).

Initial tableau:

Row	A	B	C	D
R1	a	a	a	b12
R2	b21	b22	a	a

Apply C→D: both rows agree on C (both 'a'), so equate D-values → b12 becomes **a**.

Updated tableau:

Row	A	B	C	D
R1	a	a	a	a
R2	b21	b22	a	a

Row R1 is now all "a" — matches R exactly. **Decomposition is lossless-join.**

Q7. TEACHER(TID, Subject, Hobby) — MVDs and 4NF

Sample data shows every subject paired with every hobby for a teacher (Cartesian-product blow-up) — a classic MVD symptom.

MVDs: TID →→ Subject and TID →→ Hobby (independent of each other).

Since TID is not a superkey of the whole relation, R is not in 4NF.

4NF Decomposition:

- TEACHER_SUBJECT(TID, Subject)
- TEACHER_HOBBY(TID, Hobby)

Q8. Join dependencies and 5NF

A Join Dependency JD(R1,...,Rn) holds if joining the projections always reconstructs R exactly. 5NF = every JD is implied by the candidate keys.

Example: SUPPLY(Supplier, Part, Project) — ternary relationship rule: "if S supplies P, S supplies J, and P is used in J, then S supplies P to J."

Storing this as one table forces redundant combinations and update/insertion anomalies.

5NF Decomposition:

- SUPPLIER_PART(Supplier, Part)
- SUPPLIER_PROJECT(Supplier, Project)
- PART_PROJECT(Part, Project)

Joining all three reconstructs SUPPLY exactly; no two alone can do it — this is the classic 3-way join dependency.

Q9. Poorly designed college schema — anomalies and 3NF

Schema: COLLEGE_INFO(StudentID, StudentName, CourseID, CourseName, DeptID, DeptName, HODName)

Anomalies:

- Insertion: can't add a new course until a student enrolls in it.
- Deletion: deleting the only student in a course wipes out course/department/HOD info.
- Update: changing an HOD's name requires updating every matching row — risk of inconsistency.

FDs: StudentID→StudentName,CourseID; CourseID→CourseName,DeptID; DeptID→DeptName,HODName

3NF Decomposition:

- STUDENT (StudentID, StudentName, CourseID)
- COURSE(CourseID, CourseName, DeptID)
- DEPARTMENT(DeptID, DeptName, HODName)

Q10. Closures and candidate keys

R(A,B,C,D,E,F), FDs: A→BC, B→D, CD→E, E→A

Single-attribute closures:

- $\{A\}^+ = \{A,B,C,D,E\}$ — missing F
- $\{B\}^+ = \{B,D\}$
- $\{C\}^+ = \{C\}$
- $\{E\}^+ = \{A,B,C,D,E\}$ — missing F

No single attribute reaches F (F appears in no RHS), so F must be in every candidate key.

- $\{A,F\}^+ = \{A,B,C,D,E,F\}$ = all attributes ✓
- $\{E,F\}^+ = \{A,B,C,D,E,F\}$ = all attributes ✓
- $\{B,F\}^+ = \{B,D,F\}$ ✗
- $\{C,F\}^+ = \{C,F\}$ ✗

Candidate Keys = {A,F} and {E,F} (A and E are mutually determined via A→BC→...→E and E→A, making them interchangeable in the key).